Auditing Model Substitution in LLM APIs: Why Software Tests Fail and Hardware Attestation Wins

Commercial LLM APIs can covertly swap models; empirical evidence shows software-only detection is brittle under realistic deployment noi

Reference: arXiv:2504.04715

The Model Substitution Problem: Economic Incentives Drive Covert Swaps

H0: outputs from advertised model M
H1: outputs from substitute M' (smaller/quantized/fine-tuned/other family)

Why incentives exist:
- Buyers cannot inspect deployed weights
- Cheaper inference raises margins
- Output-only audits lack execution proof

Four substitution types:
- Smaller model
- FP8 quantized variant
- Fine-tuned derivative
- Different model family

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Model family	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3.1-8B	68.4→67.9 (-0.5)	84.7→83.9 (-0.8)	52.1→50.8 (-1.3)	34.6→33.8 (-0.8)
Llama-3.1-70B	82.0→81.7 (-0.3)	93.4→92.9 (-0.5)	68.9→67.8 (-1.1)	49.2→48.7 (-0.5)
Qwen2.5-7B	70.1→69.4 (-0.7)	86.3→85.6 (-0.7)	54.8→53.2 (-1.6)	36.9→35.9 (-1.0)
Qwen2.5-32B	79.8→79.2 (-0.6)	91.5→90.8 (-0.7)	65.1→64.0 (-1.1)	46.8→45.9 (-0.9)
Mixtral-8x7B	77.4→76.8 (-0.6)	89.9→89.2 (-0.7)	61.3→60.0 (-1.3)	43.5→42.8 (-0.7)

Small benchmark drops (often within evaluation variance) make covert FP8 substitution hard to detect via outputs alone.

Higher Temperature Amplifies Performance Gaps but Increases Noise

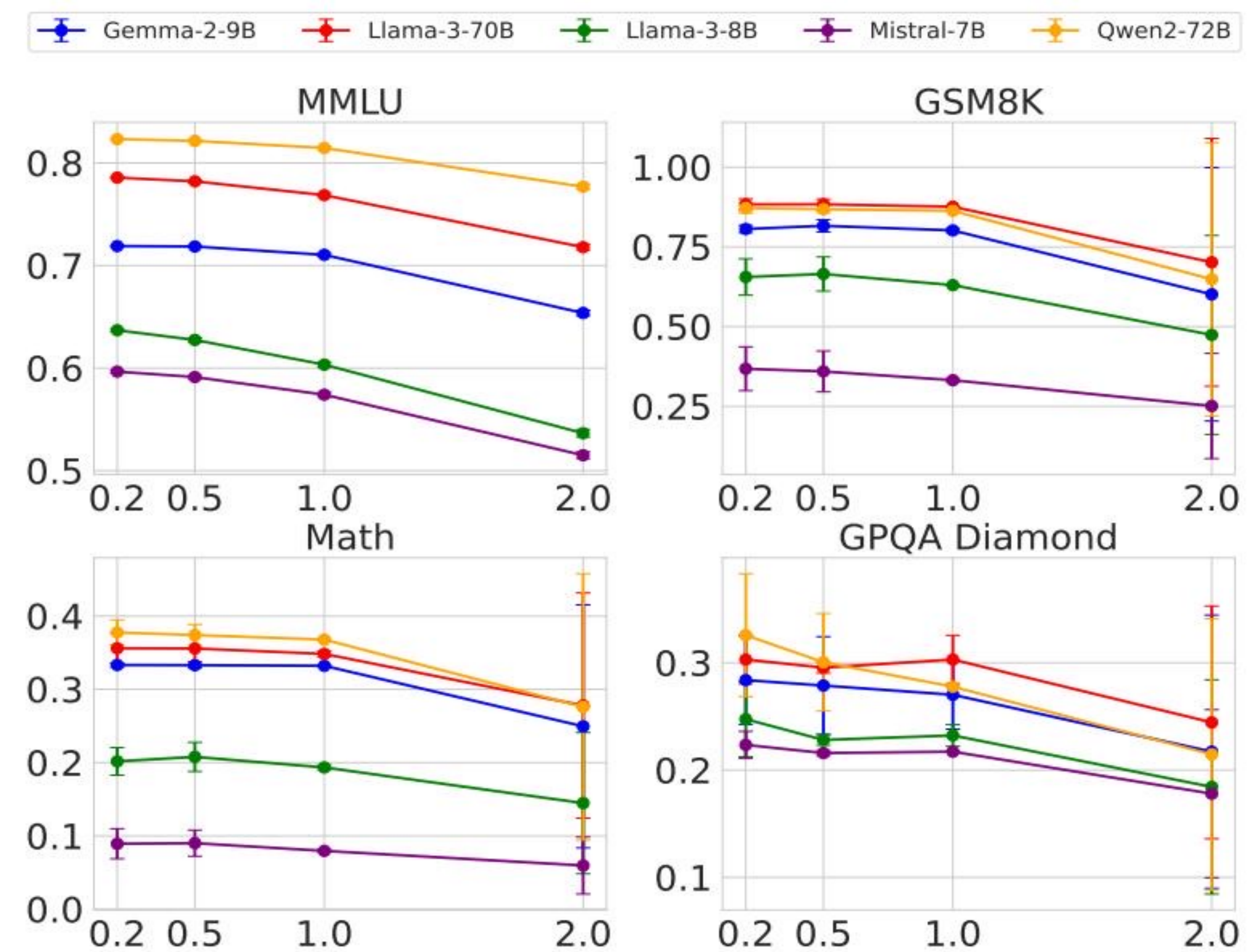

Figure 1 evidence:

- Accuracy drops as temperature rises
- At T=2.0 stochasticity spikes
- Original-substitute gap becomes noisy

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Figure 5: same Llama-3-8B, same prompts, different log-probs acros

This nondeterminism breaks stable log-prob fingerprints in product

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

$$P_{mix} = (1-p)\cdot P(M) + p\cdot P(M')$$

As $p \to 0$, mixed outputs converge to advertised distribution.

Low-rate substitution impact:
- $p=0.20$: moderate samples
- $p=0.10$: large sample budget
- $p=0.05$: expensive
- $p=0.01$: practically infeasible

Software-Only Auditing Methods Are Fundamentally Unreliable

Method type	Weakness	Practical limitation
Text-based statistical tests (MMD)	Tiny shifts for subtle substitutions	Needs very large query budgets; sensitive to decoding variance
Log-probability methods	Non-deterministic across stack/hardware	No stable production fingerprint
Adversarial countermeasures	Randomized routing + benchmark evasion	Detection delayed/bypassed; auditor cost escalates

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

TEE workflow:
1) Load signed weights in enclave
2) Remote attestation binds code+weights
3) Serve inference with integrity/confidentiality
4) Verifier checks quote before trust

Why TEEs beat software-only audits:
• Cryptographic guarantees (not probabilistic)
• Robust to randomized routing and nondeterminism
• Practical deployment via confidential computing (e.g., NVIDIA)

Key Takeaways: Close the Verification Gap with Hardware Attestation

- Model substitution is economically motivated and hard to detect via software outputs
- FP8 quantization often preserves benchmark scores closely
- Inference nondeterminism defeats log-probability fingerprinting
- Randomized substitution drives query complexity to impractical levels
- TEEs are the most actionable path to verifiable LLM API integrity

Call to action: standardize attestation-backed verification for high-stakes LLM APIs.